

Bhargavi Parthasarathy

📍 Syracuse, NY 🔗 bhargaviparthasarathy.github.io

Education

Syracuse University PhD, Mathematics Advisor: Prof. Graham J. Leuschke	Aug 2026
Syracuse University MS, Mathematics	May 2021
Indian Institute of Science Education and Research (IISER), Pune Integrated BS-MS program Master of Science in Mathematics	May 2019

Research Interests

Commutative algebra: maximal Cohen-Macaulay modules over Cohen-Macaulay rings
Formalization of mathematics for proof assistants

Publications

Homomorphisms of Maximal Cohen-Macaulay modules over the cone of the Fermat cubic (in preparation)
Homomorphisms of Maximal Cohen-Macaulay modules over the cone of all elliptic curves (in preparation)

Invited Conference and Seminar Talks

2026 Spring Eastern Sectional Meeting Homomorphisms of Maximal Cohen-Modules over the cone of an elliptic curve	Mar 2026
2026 Joint Mathematical Meetings Homomorphisms of Maximal Cohen-Modules over the cone of an elliptic curve	Jan 2026
Commutative Algebra and Algebraic Geometry Seminar, CUNY Graduate Center Homomorphisms of Maximal Cohen-Modules over the cone of an elliptic curve	Nov 2025
CAAG student seminar, CUNY Graduate Center Matrix Factorizations: An Incomplete Summary	Nov 2025
Algebra Seminar, University of Connecticut Homomorphisms of Maximal Cohen-Modules over the cone of an elliptic curve	Nov 2025
Arithmetic Geometry Seminar, Binghamton University Homomorphisms of Maximal Cohen-Modules over the cone of an elliptic curve	Nov 2025
2025 Fall Eastern Virtual Sectional Meeting Homomorphisms of Maximal Cohen-Modules over the cone of an elliptic curve	Oct 2025
Algebra Seminar, Syracuse University Homomorphisms of Maximal Cohen-Modules over the cone of an elliptic curve	Sep 2025
AWM Research Symposium, University of Wisconsin-Madison Homomorphisms of Maximal Cohen-Modules over the cone of an elliptic curve	May 2025

Awards and Grants

AWM Research Symposium Travel Grant Association for Women in Mathematics	May 2025
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AMS Sectional Travel Grant
American Mathematical Society

Departmental Summer Fellowship
Syracuse University

Oct 2023

May 2023

Teaching

Syracuse University Instructor of Record

Taught in small-section formats (≈ 25 students) as part of coordinated multi-section courses. Summer courses were taught in accelerated 6 week format with frequent assessments. Presented the course material as lectures; created exams, quizzes, and worksheets; assigned final grades.

MAT 284 Business Calculus	Spring 25, Spring 23
MAT 285 Life Sciences Calculus I	Spring 24, Summer 22, Spring 21
MAT 295 Calculus I	Fall 24, Spring 22, Fall 21
MAT 296 Calculus II	Summer 25, Summer 24, Fall 23

Recitation Leader

Conducted recitations and set quizzes; graded quizzes and exams.

MAT 183 Elements of Modern Mathematics	Fall 19, Fall 20
MAT 284 Business Calculus	Spring 20
MAT 295 Calculus I	Fall 22
MAT 296 Calculus II	Fall 25

Math Drop-in Clinic Instructor

Provided specialized support for multiple differential equations and linear algebra courses	Spring 26
Provided one-on-one support for undergraduate mathematics courses	Spring 24, Fall 21, Summer 21

Grader

MAT 485 Differential Equations and Matrix Algebra for Engineers	Spring 26
MAT 121 Probability and Statistics for Liberal Arts I	Summer 20

Indian Institute of Science Education and Research (Pune)

Teaching Fellow

Mentored students for their semester project using the history of science, technology and medicine framework.

HSS 201 Introduction to History of Science, Medicine and Technology	Fall 18
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Contributed Conference and Seminar Talks

Working ALgebra Knowledge Seminar (WALKS), Syracuse University Maximal Cohen-Macaulay modules and the elliptic group law	Sep 2025
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50th Annual New York State Regional Graduate Mathematics Conference (ANYSRGMC), Syracuse University Homomorphisms of Maximal Cohen-Macaulay Modules over the cone of an elliptic curve	Apr 2025
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Working ALgebra Knowledge Seminar (WALKS), Syracuse University Matrix Factorizations: An Incomplete Summary	Jan 2025
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Working ALgebra Knowledge Seminar (WALKS), Syracuse University AR Sequences	Oct 2023
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MGO Graduate Colloquium, Syracuse University AR sequences and Cohen-Macaulay modules	Apr 2023
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Departmental Service

Directed Reading Program, Syracuse University

Graduate Mentor

Had weekly meetings with four undergraduate students to discuss a dive deeper into a mathematical topic of their choice which culminated with a presentation given by them at the end of the semester.

- Spring 2022 - Group theory (Yuming Jiang)
- Fall 2023 - Graph Colorings and the Axiom of Choice (Lillian Impastato)
- Spring 2025 - Applications of Gröbner basis (Qiong Wu and Meagan Gonzalez)

Syracuse Mathematics Peer Mentoring Program

2023-24

Peer Mentor

Organized various social events for first year graduate students to foster a supportive environment and create a sense of community.

Mathematics Graduate Organization, Syracuse University

Colloquium Organizer

2021-22

Organized weekly research and expository talks given by graduate students.

Association for Women in Mathematics-Syracuse University Chapter

President

2021-22

Vice President

2020-21

Organized job preparation workshops, mental health discussions, academic and industry career panels, research communication event, team building exercises and movie screenings for the mathematics graduate student body.

Outreach

Networking 101 at the Conference for Undergraduate women and Gender Minorities in Physics (CU*IP)

Jan 2025

Facilitated a group discussion among Physics undergraduates (women and gender minorities) about constructing elevator pitches and talking to professors.

Math E-Carnival with Institute for Advanced Study's Women and Math program

Apr 2021

Presented various activities to an audience of children and interacted with the underlying mathematics concepts in these activities

Science and Technology Entry Program (STEP) at Syracuse University

Sep 2020-Apr 2021

Worked with middle and high school students on a weekly basis to improve their understanding of various mathematical concepts.

Professional Development

Teaching Seminar

Fall 2024

Department of Mathematics, Syracuse University

A weekly seminar to discuss pedagogy with the mathematics education experts in the department and testing different ways to implement them.

Future Professionals Program

2023-25

Women in Science and Engineering (WISE), Syracuse University

A two year long professional development program with events that address various aspects of building a career in STEM including networking, preparing for academic jobs, improving instructional abilities.

Workshops Attended

Commutative Algebra and its Interaction with Algebraic Geometry

May 2023

CMND, Notre Dame and SLMATH, Berkeley

Women and Mathematics: Representation theory and categorification

Apr 2021

Institute of Advanced Study, Princeton

Conferences Attended

AWM Research Symposium, University of Wisconsin-Madison	May 2025
50th ANYSRGMC, Syracuse University	Mar 2025
Route 81, Cornell University	Nov 2024
AMS Fall Eastern Sectional Meeting, University at Albany	Oct 2024
49th ANYSRGMC, Syracuse University	Apr 2024
Fairfax Algebra Days, George Mason University	Mar 2024
Introductory Workshop: Commutative Algebra, SLMath, Berkeley	Jan 2024
Connections Workshop: Commutative Algebra, SLMath, Berkeley	Jan 2024
Route 81, Syracuse University	Nov 2023
AMS Fall Central Sectional Meeting, Creighton University	Oct 2023
KUMUNU, University of Missouri	Sep 2023
Morgantown Algebra Days, West Virginia University	Apr 2023
48th ANYSRGMC, Syracuse University	Apr 2023
47th ANYSRGMC, Syracuse University	Apr 2022
46th ANYSRGMC, Syracuse University	Apr 2021
45th ANYSRGMC, Syracuse University	Apr 2020